

PARTS LIST

Semi-conductors:

- IC1 - LM386 low-power audio amplifier
 IC2 - LM7809, 9V voltage regulator
 LED1 - 5mm LED
 BR1 - 1A bridge rectifier module
 D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 270-kilo-ohm
 R2 - 120-kilo-ohm
 R3 - 64-kilo-ohm
 R4 - 33-kilo-ohm
 R5 - 15-kilo-ohm
 R6 - 8.2-kilo-ohm
 R7 - 3.9-kilo-ohm
 R8 - 2-kilo-ohm
 R9 - 680-ohm
 R10 - 10-ohm
 R11 - 180-ohm
 R12 - 20-ohm
 R13 - 100-ohm, 1W
 VR1 - 220-ohm potentiometer

Capacitors:

- C1-C4, C15 - 100nF ceramic disk
 C5 - 1000 μ F, 25V electrolytic
 C6, C8 - 330nF ceramic disk
 C7 - 100 μ F, 25V electrolytic
 C9 - 22 μ F, 25V electrolytic
 C10 - 470 μ F, 25V electrolytic
 C11 - 33nF ceramic disk
 C12 - 47 μ F, 25V electrolytic
 C13 - 1 μ F, 25V electrolytic
 C14 - 10nF ceramic disk
 C16 - 0.1 μ F, 400V polyester

Miscellaneous:

- S1, S2 - On/off switch
 CON1, CON3 - 2-pin terminal connector
 CON4 - 9-pin connector
 CON2, CON5 - 2-pin connector
 CON7 - 230V AC primary to 12V AC, 500mA secondary transformer
 F1 - 200mA fuse

the circuit outputs around 0.5W. The loudspeaker can be 4-ohm, but 8-ohm or higher is preferred to avoid overloading LM386.

The power supply is built around voltage regulator 7809 which gives a

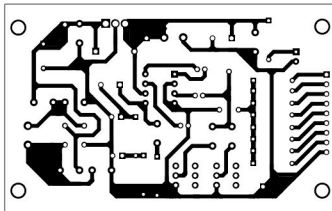


Fig. 2: An actual-size PCB layout for the circuit

